

Digital Signatures

RSA-PSS signing, tamper detection, X.509 identity binding, and trust.

01 IMPLEMENTATION OVERVIEW

This lab creates a fresh 2048-bit RSA signing key pair, signs a fictional DeRexi policy message using **RSA-PSS with SHA-256**, verifies the original message, and proves that a one-phrase change causes verification to fail. It uses only fictional data and throwaway demo keys.

CORE SIGNING + VERIFICATION

```
private_key = rsa.generate_private_key(
    public_exponent=65537, key_size=2048)

signature = private_key.sign(
    message, pss_padding(), hashes.SHA256())

public_key.verify(
    signature, message, pss_padding(), hashes.SHA256())
```

01 Signing key pair

2048-bit RSA with public exponent 65537. The private key creates signatures and remains secret; the public key verifies signatures and may be distributed.

02 RSA-PSS + SHA-256

The message is hashed with SHA-256 and signed with randomized RSA-PSS padding. Signing the same message twice produces different signatures that both verify.

03 Tamper detection

The original signature verifies successfully. A changed phrase produces a different digest, so verification fails and the lab reports **CHANGE DETECTED**.

02 SECURITY GUARANTEES

Integrity

A valid signature confirms that the signed bytes were not altered after signing.

Authenticity

Verification proves the signature was created by the holder of the matching private key.

Confidentiality?

No. A digital signature does not encrypt the message. The content remains readable.

03 FILE-SIGNATURE TEST

The lab also signs a fictional policy file. The unchanged file verifies as **PASS**; after one wording change, verification fails. The file is restored and deleted at the end of the run, preventing test artifacts from remaining in the repository.

VERIFICATION: Original signature: **PASS** | Tampered message: **CHANGE DETECTED** | Modified file: **CHANGE DETECTED**

04 X.509 CERTIFICATE + KEY MANAGEMENT

Certificate identity

Lifecycle controls

Private-key handling

Trust model

05 EXECUTION EVIDENCE

06 RUBRIC COVERAGE

01 Signature system

02 Certificate + trust

03 Safety

2 of 2